

	Infrastructure	Operation Domains		
HPC	Layer 1: Hardware Centric Security (Device/Edge/ Cloud/HPC)	Layer 6: Network infrastructure and perimeter security		
		Layer 5: Data Centric Security		
Cloud		Layer 2: Software Centric Security (Middleware/OS)	Layer 3: Application Centric Security (Apps/Services)	Layer 4: User Centric Security (Authentication Access/Identity / Orchestration)
Edge				
Devices				
Cross Layer 1: Automated security processes and Cyber-Resilient AI Lifecycle				
Cross Layer 2: Dynamic Risk Management & Operational Cyber Resilience				
Cross Layer 3: Interoperable Security Mechanisms				
Cross Layer 4: Agile Governance, Compliance-by-Design & Automated Assurance				

Fig. 1: Security Framework with separated security verticals in dedicated boxes.

Security Framework with Separated Security Verticals

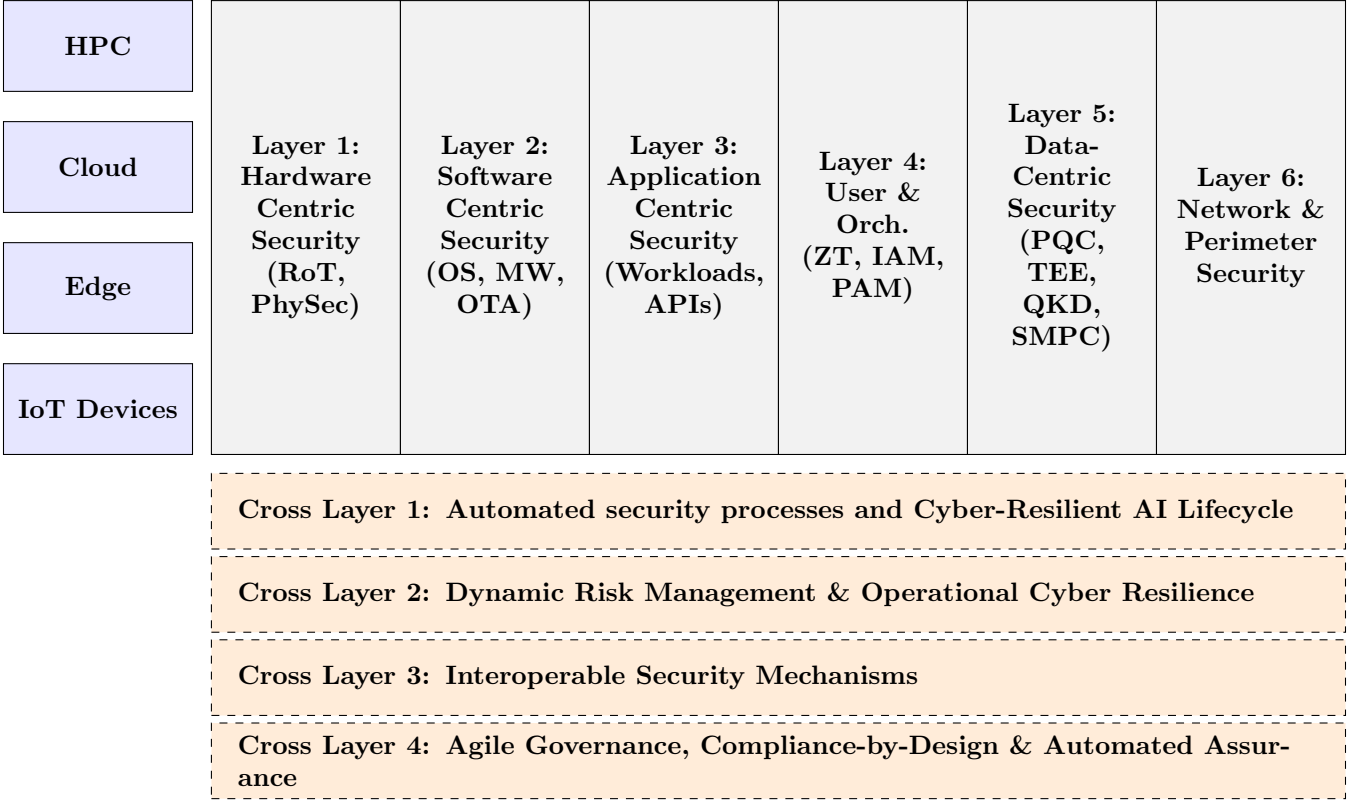


Fig. 2: Security framework with L1–L6 security layers and four cross-cutting capability domains.

Execution Environments		
HPC	Layer 1: Hardware Centric Security (Device/Edge/ Cloud/HPC)	Layer 2: Software Centric Security
Cloud		
Edge		
IoT		
	Cross Layer 1: Automated security processes and Cyber-Resilient AI Lifecycle	
	Cross Layer 2: Dynamic Risk Management & Operational Cyber Resilience	
	Cross Layer 3: Interoperable Security Mechanisms	
	Cross Layer 4: Agile Governance, Compliance-by-Design & Automated Assurance	